Qinfeng Zhu

Ph.D. Candidate in Computer Science, University of Liverpool
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Seeking 2026 research internships and 2027 full-time research roles in embodied intelligence.

Education

Duke University (Kunshan) Sep 2026 – present
Visiting Researcher in Embodied Intelligence; supervised by Prof. Kaizhu Huang

University of Liverpool Sep 2023 – Aug 2027 (expected)
Ph.D. in Computer Science; based at Xi'an Jiaotong-Liverpool University (XJTLU); advised by Dr. Lei Fan and Dr. Anh Nguyen

University of Liverpool Sep 2021 – Mar 2023
Master of Research in Computer Science; based at XJTLU; advised by Dr. Ming Xu

Jiangsu University of Science and Technology Sep 2017 – Jun 2021
Bachelor's Degree in Internet of Things Engineering

Research Experience

1. Embodied Intelligence *(Ongoing Research Direction)*

- **Agentic robotics safety – LEASH:** A lease-enforced safety harness with task, action and commit gates sharing one safety policy. RoboRisk evaluates safety across 92 simulation and 13 real-robot tasks. [1] Video demo
- **Humanoid robot collaboration *(Nov 2026 – present)*:** Ongoing research based on the Unitree G1; technical details are not yet public.

2. Panoramic and Spherical Vision

- **Sphere-native perception:** Study intrinsic spherical features for rotation-robust panoramic dense prediction, alongside a survey organizing scene-understanding methods from distortion-aware engineering to sphere-native modeling. [3, 4]
- **Full-sphere tracking:** Investigate person tracking from both fixed and moving panoramic cameras, extending the current research focus from dense scene prediction to dynamic full-sphere perception. [2]

3. 3D and 2D Vision

- **3D scene understanding:** Point-cloud data augmentation survey and co-first-author work on SemanticUrban for urban semantic understanding. [5, 7]
- **2D and multispectral perception:** Category-specific fusion, Vision Mamba, local-global modeling and pre-training, with applications in remote sensing, indoor segmentation, environmental monitoring and geospatial risk modeling. [6, 8–13]

Academic Service

Conference reviewer: ICLR, AAAI, IJCAI, NeurIPS, WACV.

Journal reviewer: TNNLS, TCSVT, T-ITS, TNSE, ISPRS JPRS, PR, NN, ESWA, EAAI, Neurocomputing.

Selected Publications

[1] **LEASH: Stay Safe in Agentic Robotics**
Qinfeng Zhu, Weiguang Zhao, Ke Sun, Xihao Guo, Yunxi Jiang, Rui Zhang, Lei Fan, Kaizhu Huang.
arXiv, 2026. Under review. Video demo.

[2] **PanoPed: Beyond Bounding Boxes for Sim-to-Real Panoramic Pedestrian Tracking**
Qinfeng Zhu, Weiguang Zhao, Yunxi Jiang, Anh Nguyen, Lei Fan.
arXiv, 2026. Under review.

[3] **SO3UFormer: Learning Intrinsic Spherical Features for Rotation-Robust Panoramic Dense Prediction**
Qinfeng Zhu, Yunxi Jiang, Lei Fan.
arXiv, 2026. Under review at International Journal of Computer Vision (IJCV).

[4] **Panoramic Scene Understanding: A Survey from Distortion-Aware Engineering to Sphere-Native Modeling**
Qinfeng Zhu, Lei Fan.
arXiv, 2026. Under review at International Journal of Computer Vision (IJCV).

[5] **Advancements in Point Cloud Data Augmentation for Deep Learning: A Survey**
Qinfeng Zhu, Lei Fan, Ningxin Weng.
Pattern Recognition (PR), 2024.

[6] **ClassWise-CRF: Category-Specific Fusion for Enhanced Semantic Segmentation of Remote Sensing Imagery**
Qinfeng Zhu, Yunxi Jiang, Lei Fan.
Neural Networks (NN), 2026.

[7] **Towards Accurate Urban Scene Understanding Using Point Clouds: The SemanticUrban Dataset**
Yuan Fang, **Qinfeng Zhu** (equal), Yuanzhi Cai, Lei Fan.
Expert Systems with Applications (ESWA), 2026.

[8] **Rethinking Scanning Strategies with Vision Mamba in Semantic Segmentation of Remote Sensing Imagery**
Qinfeng Zhu, Yuan Fang, Yuanzhi Cai, Cheng Chen, Lei Fan.
IEEE JSTARS, 2024.

[9] **SwinMamba: A Hybrid Local-Global Mamba Framework for Enhancing Semantic Segmentation of Remotely Sensed Images**
Qinfeng Zhu, Han Li, Liang He, Lei Fan.
Digital Signal Processing, 2026.

[10] **IndoorMS: A Multispectral Dataset for Semantic Segmentation in Indoor Scene Understanding**
Qinfeng Zhu, Jingjing Xiao, Lei Fan.
IEEE Sensors Journal, 2025.

[11] **Enhancing Environmental Monitoring Through Multispectral Imaging: The WasteMS Dataset for Semantic Segmentation of Lakeside Waste**
Qinfeng Zhu, Ningxin Weng, Lei Fan, Yuanzhi Cai.
MMM, 2025.

[12] **A Generalised Pre-Training Strategy for Deep Learning Networks in Semantic Segmentation of Remotely Sensed Images**
Yuan Fang, Yuanzhi Cai, Jagannath Aryal, **Qinfeng Zhu**, Hong Huang, Cheng Zhang, Lei Fan.
CGI / LNCS, 2026.

[13] **From Landslide Conditioning Factors to Satellite Embeddings: Evaluating the Utilisation of Google AlphaEarth for Landslide Susceptibility Mapping Using Deep Learning**
Yusen Cheng, **Qinfeng Zhu**, Lei Fan.
arXiv, 2026. Under review.

朱钦峰（Qinfeng Zhu）

利物浦大学计算机科学博士研究生

邮箱：zhuqinfeng1999@gmail.com | 主页：zhuqinfeng1999.github.io

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寻求具身智能方向的 *2026* 年研究实习，以及 *2027* 年毕业后的学术界或业界研究岗位。

教育背景

杜克大学（昆山） 2026.09 – 至今

具身智能方向访问研究；导师：*Prof. Kaizhu Huang*

利物浦大学（University of Liverpool） 2023.09 – 2027.08（预计）

计算机科学博士研究生；在西交利物浦大学（*XJTLU*）开展研究；导师：*Dr. Lei Fan*、*Dr. Anh Nguyen*

利物浦大学（University of Liverpool） 2021.09 – 2023.03

计算机科学研究型硕士（*MRes*）；在西交利物浦大学开展研究；导师：*Dr. Ming Xu*

江苏科技大学 2017.09 – 2021.06

物联网工程学士

研究经历

1. 具身智能（持续研究方向，Ongoing）

- **Agentic Robotics 安全 – LEASH**：通过任务、动作与执行提交三层安全门，将共享安全策略贯穿智能体规划与物理执行。RoboRisk 在 92 个仿真任务与 13 个真实机器人任务中评估安全性。[1] 视频演示
- **人形机器人协作（2026.11 – 至今）**：基于宇树 Unitree G1 开展研究；具体技术路线暂不公开。

2. 全景与球面视觉

- **球面原生感知**：研究面向旋转鲁棒全景稠密预测的内禀球面特征，并系统梳理从畸变感知工程到球面原生建模的场景理解方法。[3, 4]
- **全景行人跟踪**：研究固定与移动全景相机下的行人跟踪，将稠密场景预测扩展到动态全景感知。[2]

3. 3D 与 2D 视觉

- **三维场景理解**：点云数据增强综述；共同第一作者工作 SemanticUrban，面向城市点云语义理解。[5, 7]
- **二维与多光谱感知**：类别特定融合、Vision Mamba、局部与全局建模和预训练；应用于遥感、室内分割、环境监测及地理风险建模。[6, 8–13]

学术服务

会议审稿： ICLR, AAAI, IJCAI, NeurIPS, WACV.

期刊审稿： TNNLS, TCSVT, T-ITS, TNSE, ISPRS JPRS, PR, NN, ESWA, EAAI, Neurocomputing.

代表性论文（论文标题保留英文原文）

[1] **LEASH: Stay Safe in Agentic Robotics**
Qinfeng Zhu, Weiguang Zhao, Ke Sun, Xihao Guo, Yunxi Jiang, Rui Zhang, Lei Fan, Kaizhu Huang.
arXiv, 2026. Under review. 视频演示。

[2] **PanoPed: Beyond Bounding Boxes for Sim-to-Real Panoramic Pedestrian Tracking**
Qinfeng Zhu, Weiguang Zhao, Yunxi Jiang, Anh Nguyen, Lei Fan.
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[13] **From Landslide Conditioning Factors to Satellite Embeddings: Evaluating the Utilisation of Google AlphaEarth for Landslide Susceptibility Mapping Using Deep Learning**
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